

RAKESH KUMAR

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Education

Indian Institute of Technology, Patna
B.Tech in Metallurgical and Materials Engineering

7.01 CPI (Till Sem-5)
Aug 2023 – May 2027

Projects

Limit Order Booking

Github

- Built a high-performance C++ limit order book and matching engine simulating real-world trading systems with price-time (FIFO) priority.
- Designed optimized data structures (maps, lists, hash maps) to enable efficient order matching, fast cancellations, and scalable performance under heavy load.
- Implemented robust failure handling and validation mechanisms, including partial fills, orphan order cleanup, and stress testing with 100K+ operations to ensure system integrity.

MFCC Feature Extraction Pipeline for Audio Processing

Github

- Built an end-to-end MFCC feature extraction pipeline from scratch, implementing signal processing stages including pre-emphasis, FFT, Mel filter banks, and DCT.
- Designed a modular and testable system architecture, enabling independent validation of each stage and improving debugging of intermediate audio transformations.
- Addressed key failure modes such as numerical instability ($\log(0)$), spectral leakage, and filter misalignment, ensuring robust and accurate feature generation.

Experience

Event Management Coordinator — Materials Engineering Society (MatES)

- Served as Coordinator of the Event Management Committee, leading planning and execution of academic and outreach events.
- Hosted and conducted multiple quizzes during the inaugural “Material Science Quiz and Exhibition,” ensuring smooth event flow and audience engagement.
- Managed end-to-end event operations including coordination, scheduling, and participant handling for 200+ students across schools in Patna.
- Collaborated with cross-functional teams to drive participation, enhance event visibility, and deliver a structured and impactful experience.

Accomplishments

- Reached 1300+ rating on Codeforces by consistently solving algorithmic problems and competing in rated contests.
- Secured a place among the Top 1.2 percent of the 1.4 million+ candidates who appeared for IIT-JEE.

Skills

Programming Languages: C, C++, Python, MATLAB(Basics)

Softwares: OriginPro, ImageJ, MATLAB, MS Word, MS Powerpoint, MS Excel, VS Code, Canva, LaTeX.

Soft Skills: Leadership, Time and Work Management, Problem Solving and Presentation Skill

Leadership/ Extra-Curricular Activities

- Serving as a Coordinator of Materials Engineering Society (MatES), IIT Patna 2025 - Present
- Keen interest in strategic and competitive sports such as Volleyball, cricket and Football.